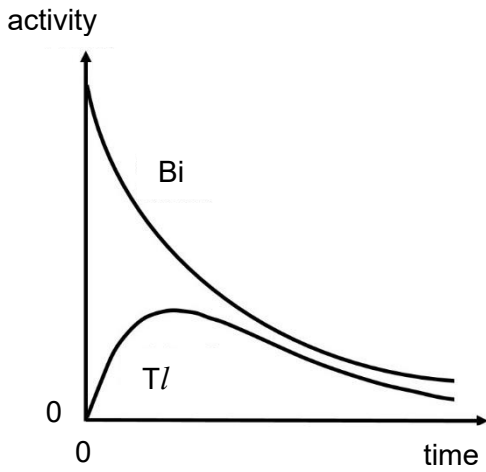


	highest wavelength of 700 nm	
9(a)	The half-life of a radioactive nuclide is the <u>time taken for the radioactive nuclei to reduce to half its initial value.</u>	[1]
9(b)(i)	For α -decay: ${}_{83}^{214}\text{Bi} \rightarrow {}_{81}^{210}\text{Tl} + {}_2^4\text{He}$	[1]
	For β^- -decay: ${}_{83}^{214}\text{Bi} \rightarrow {}_{84}^{214}\text{Po} + {}_{-1}^0\text{e}$	[1]
9(b)(ii)	mass difference = $213.9987u - (209.9901u + 4.0015u)$ = $0.0071u$ energy released = Δmc^2 = $0.0071(1.66 \times 10^{-27})(3.00 \times 10^8)^2$ = 1.06074×10^{-12} $\approx 1.06 \times 10^{-12} \text{ J}$	[1] [1] for correct substitution [1] for correct answer

No.	Solution	Remarks
9(b)(iii)	The <u>mass difference of reactant and products</u> in both decay processes are <u>different</u>. So the total <u>amount of energy released for each process will also be different</u>. Hence the γ-ray photons emitted will have different energies.	[1] [1]
9(c)(i) and 9(c)(ii)		[1] for correct Bi graph [1] for correct Tl graph
9(d)(i)	214 g of Bi contains 1 mol of atoms, which is 6.02×10^{23} atoms. number of atoms in sample $= \frac{2.0 \times 10^{-6}}{214} (6.02 \times 10^{23})$ $= 5.626 \times 10^{15}$ $\approx 5.6 \times 10^{15} \text{ (shown)}$	[1] correct substitution
9(d)(ii)	activity $A = \lambda N$ $\lambda = \frac{A}{N}$ $= \frac{3.3 \times 10^{12}}{5.626 \times 10^{15}}$ $= 5.866 \times 10^{-4}$ $\approx 5.9 \times 10^{-4} \text{ s}^{-1}$	[1] correct substitution [1] correct answer
9(d)(iii)	$t_{1/2} = \frac{\ln 2}{\lambda}$ $= \frac{\ln 2}{5.866 \times 10^{-4}}$ $= 1182$ $\approx 1.2 \times 10^3 \text{ s}$	[1] for correct answer

No.	Solution	Remarks
9(d)(iv)	$A = A_0 e^{-\lambda t}$ $3.3 \times 10^6 = 3.3 \times 10^{12} e^{-5.866 \times 10^{-4} t}$ $\frac{3.3 \times 10^6}{3.3 \times 10^{12}} = e^{-5.866 \times 10^{-4} t}$ $t = 2.355 \times 10^4$ $\approx 2.4 \times 10^4 \text{ s}$	[1] correct substitution [1] correct answer
9(e)	<u>β - particles are fast moving electrons</u> and the β - particles are emitted collide with the atoms of the lead container. These collisions result in the β - particles undergoing <u>deceleration</u>, either being slowed down <u>by collision or by deflection</u>. The electrons will then <u>release energy</u> in the form of the continuous <u>spectrum of the X-ray produced</u>, also called Bremsstrahlung radiation.	[1] [1] [1] [1]